Supplementary Information

Role of neutrophil-lymphocyte ratio in the prognosis of acute ischemic stroke after reperfusion therapy: A systematic review and meta-analysis

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Considering the MJA Analysis, a score of 1 indicated “yes” and 0 indicated “no” or “not described”. Additionally, a double-blind got a score of 1 and a single-blind got 0.5. The quality assessment was: low quality (0–3 points) and high quality (4–8 points) levels. Where the score was intermediate, this was classified as such. **4**

Considering the funding bias: 0 indicated “a low potential for bias”, 1 indicated “conflicts of interest declared related to industry outside of the current publication”, 2 indicated that “the study was funded by industry”, and 3 indicated “a high potential for bias”. **4**

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1. Search Strategy

1.1. Embase

1. Stroke or Stroke care or acute stroke or Cerebrovascular Accident or Brain ischemia or ischemic stroke or acute ischemic stroke or anterior circulation stroke or MCA Stroke or ICA Stroke or middle cerebral artery occlusion or MCA occlusion or internal carotid artery occlusion or ICA occlusion or large vessel occlusion or LVO
2. NLR or Neutrophil Lymphocyte Ratio or Lymphocyte Neutrophil Ratio or Neutrophil:Lymphocyte or Lymphocyte:Neutrophil
3. Reperfusion or Reperfusion therapy or endovascular thrombectomy or endovascular treatment or endovascular therapy or acute stroke treatment or acute stroke intervention or thrombolysis or tPA or transplasminogen activator or thrombectomy or EVT or Clot retrieval or mechanical thrombectomy
4. 1 AND 2 AND 3
5. Limit 4 to 'Human'
6. Limit 5 to 'English Language'
7. Limit 6 to '(article-in-press status or embase status or in-process status)'

1.2. PubMed/Medline

1. "stroke"[Mesh] OR Brain Ischemia[Title/Abstract] OR Brain infarction[Title/Abstract] OR cerebral infarction[Title/Abstract] OR Cerebrovascular Accident[Title/Abstract] OR acute ischemic stroke[All Fields] OR Title/Abstract[All Fields] OR internal carotid artery occlusion[Title/Abstract] OR large vessel occlusion[Title/Abstract] OR middle cerebral artery occlusion[Title/Abstract]
2. "neutrophil lymphocyte ratio"[All Fields] OR "NLR"[All Fields] or neutrophil lymphocyte ratio or NLR
3. "Reperfusion"[MeSH Terms] OR Reperfusion therapy[All Fields] OR endovascular thrombectomy[All Fields] OR endovascular treatment[All Fields] OR endovascular therapy[All Fields] OR thrombectomy[All Fields] OR EVT[All Fields] OR Clot retrieval[All Fields] OR mechanical thrombectomy[All Fields] OR thrombolysis[All Fields] or tPA[All Fields] or transplasminogen activator[All Fields] OR IVT[All Fields]
4. 1 AND 2 AND 3
5. Limit 4 to 'Human'
6. Limit 5 to 'English Language'

1.3. Cochrane

1. "Stroke" or "Stroke care" or "acute stroke" or "Cerebrovascular Accident" or "Brain ischemia" or "ischemic stroke" or "acute ischemic stroke" or "anterior circulation stroke" or "MCA Stroke" or "ICA Stroke" or "middle cerebral artery occlusion" or "MCA occlusion" or "internal carotid artery occlusion" or "ICA occlusion" or "large vessel occlusion" or LVO
2. "NLR" or "Neutrophil Lymphocyte Ratio" or "Lymphocyte Neutrophil Ratio"
3. "Reperfusion" or "Reperfusion therapy" or "endovascular thrombectomy" or "endovascular treatment" or "endovascular therapy" or "acute stroke treatment" or "acute stroke intervention" or thrombolysis or tPA or "transplasminogen activator" or thrombectomy or EVT or "Clot retrieval" or "mechanical thrombectomy"
4. 1 AND 2 AND 3

2. Supplemental Table 1: Summary of methodological quality and funding bias assessment of studies included in the meta-analysis

Study ID	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**13**	**14**	**15**	**16**	**17**	**18**
Was the study randomised?	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Was the method of randomisation appropriate (not specified = 0)	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Was the study described as being blinded?	0.5	0.5	0	0	0	0.5	0.5	0.5	0	0.5	0.5	0	0	0.5	0	0	0	0
Was the method of blinding appropriate?	1	1	0	0	0	1	1	1	0	1	1	0	0	1	0	0	0	0
Was there a description of withdrawals and dropouts?	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Was there a clear description of the inclusion/exclusion criteria?	1	0	1	1	1	1	1	1	1	1	1	1	0	1	1	1	1	1
Was the method used to assess adverse events described?	1	1	1	1	1	0	1	0	1	1	1	0	1	1	1	1	1	1
Was the method of statistical analysis described?	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
TOTAL MJA SCORE	4.5	3.5	3	3	3	3.5	4.5	3.5	3	4.5	4.5	2	2	4.5	3	3	3	3
MJA Quality Assessment	High	Int	Low	Low	Low	Int	High	Int	Low	High	High	Low	Low	High	Low	Low	Low	Low
Funding Bias Score	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Study ID	**19**	**20**	**21**	**22**	**23**	**24**	**25**	**26**	**27**	**28**	**29**	**30**	**31**	**32**	**33**	**34**	**35**	
Was the study randomised?	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
Was the method of randomisation appropriate (not specified = 0)	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
Was the study described as being blinded?	0	0	0	0	0.5	0	0	0	0	0.5	0	0.5	0.5	0	0.5	0	0	
Was the method of blinding appropriate?	0	0	0	0	1	0	0	0	0	1	0	1	1	0	1	0	0	
Was there a description of withdrawals and dropouts?	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
Was there a clear description of the inclusion/exclusion criteria?	1	0	1	1	1	1	1	1	1	1	1	0	1	1	1	1	1	

Was the method used to assess adverse events described?	0	1	1	1	1	1	1	0	1	1	1	1	1	0	1	1	1
Was the method of statistical analysis described?	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
TOTAL MJA SCORE	2	2	3	3	4.5	3	3	2	3	4.5	3	3.5	4.5	2	4.5	3	3
MJA Quality Assessment	Low	Low	Low	Low	High	Low	Low	Low	Low	High	Low	Int	High	Low	High	Low	Low
Funding Bias Score	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0

Considering the MJA Analysis, a score of 1 indicated “yes” and 0 indicated “no” or “not described”. Additionally, a double-blind got a score of 1 and a single-blind got 0.5. The quality assessment was: low quality (0–3 points) and high quality (4–8 points) levels. Where the score was intermediate, this was classified as such.

Considering the funding bias: 0 indicated “a low potential for bias”, 1 indicated “conflicts of interest declared related to industry outside of the current publication”, 2 indicated that “the study was funded by industry”, and 3 indicated “a high potential for bias”.

Abbreviations: MJA=Modified Jadad Analysis, Int=Intermediate

3. Supplemental Table 2: Eggers Test Analyses

Outcome	*Time Point*	*T Value*	*P Value*
Good Functional Outcomes	Admission	-1.57	0.584
	Delayed	-2.73	0.078
mRS 0-1	Admission	-0.83	0.936
Mortality	Admission	-0.56	0.361
	Delayed	0.88	0.724
sICH	Admission	-2.64	0.008
	Delayed	1.19	0.211
ICH	Admission	-4.77	0.001
	Delayed	-0.23	0.205
ENI	Admission	-19.06	0.786
SAI	Admission	7.10	0.071

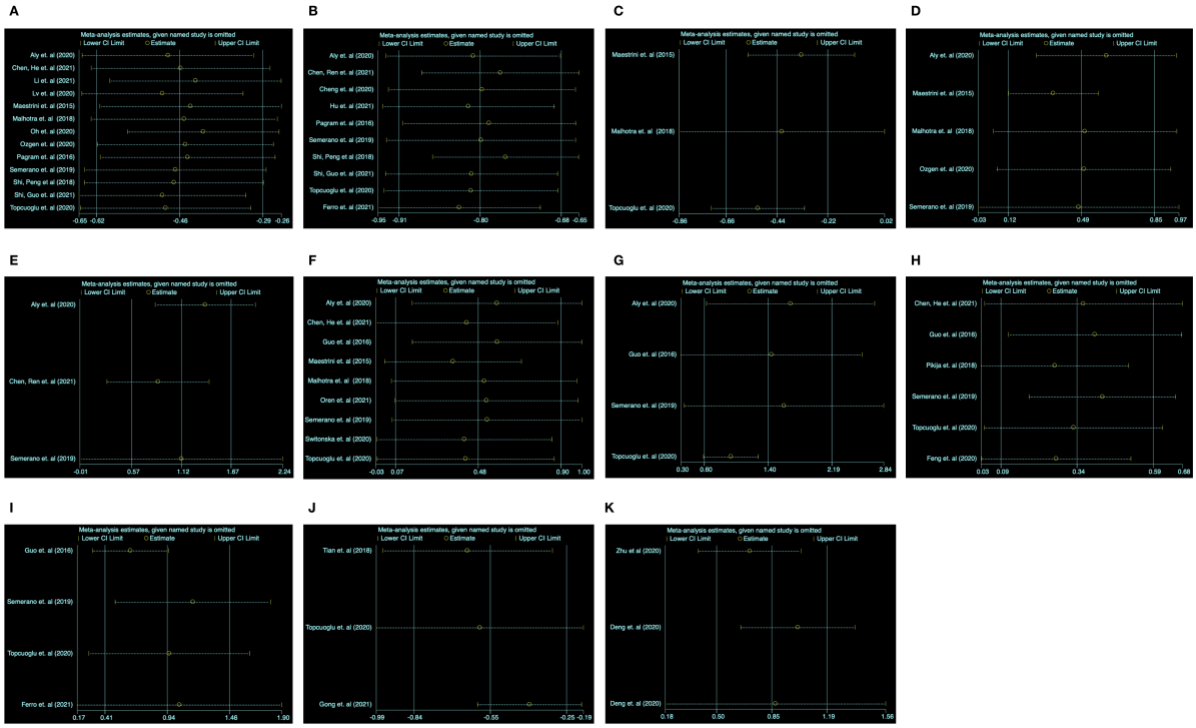
Abbreviations: mRS=Modified Rankin Scale; sICH=Symptomatic intracerebral haemorrhage; ICH=intracerebral haemorrhage; SAI=Stroke Associated Infection

4. Supplemental Table 3: Standards for Reporting Diagnostic Accuracy (STARD)-2015 Checklist

Section & Topic	No	Item	Reported on page #
TITLE OR ABSTRACT			
	1	Identification as a study of diagnostic accuracy using at least one measure of accuracy (such as sensitivity, specificity, predictive values, or AUC)	2
ABSTRACT			
	2	Structured summary of study design, methods, results, and conclusions (for specific guidance, see STARD for Abstracts)	1
INTRODUCTION			
	3	Scientific and clinical background, including the intended use and clinical role of the index test	2
	4	Study objectives and hypotheses	3
METHODS			
Study design	**5**	Whether data collection was planned before the index test and reference standard were performed (prospective study) or after (retrospective study)	3
Participants	**6**	Eligibility criteria	4
	7	On what basis potentially eligible participants were identified (such as symptoms, results from previous tests, inclusion in registry)	4
	8	Where and when potentially eligible participants were identified (setting, location and dates)	3/4
	9	Whether participants formed a consecutive, random or convenience series	4
Test methods	**10a**	Index test, in sufficient detail to allow replication	5/6
	10b	Reference standard, in sufficient detail to allow replication	5/6
	11	Rationale for choosing the reference standard (if alternatives exist)	5/6
	12a	Definition of and rationale for test positivity cut-offs or result categories of the index test, distinguishing pre-specified from exploratory	5/6
	12b	Definition of and rationale for test positivity cut-offs or result categories of the reference standard, distinguishing pre-specified from exploratory	5/6
	13a	Whether clinical information and reference standard results were available to the performers/readers of the index test	5/6
	13b	Whether clinical information and index test results were available to the assessors of the reference standard	5/6
Analysis	**14**	Methods for estimating or comparing measures of diagnostic accuracy	5/6
	15	How indeterminate index test or reference standard results were handled	5/6
	16	How missing data on the index test and reference standard were handled	5/6
	17	Any analyses of variability in diagnostic accuracy, distinguishing pre-specified from exploratory	5/6
	18	Intended sample size and how it was determined	5/6
RESULTS			
Participants	**19**	Flow of participants, using a diagram	Figure 1
	20	Baseline demographic and clinical characteristics of participants	6/Tables 1-3
	21a	Distribution of severity of disease in those with the target condition	Tables 1-3
	21b	Distribution of alternative diagnoses in those without the target condition	Tables 1-3
	22	Time interval and any clinical interventions between index test and reference standard	Tables 2-3
Test results	**23**	Cross tabulation of the index test results (or their distribution) by the results of the reference standard	6-13

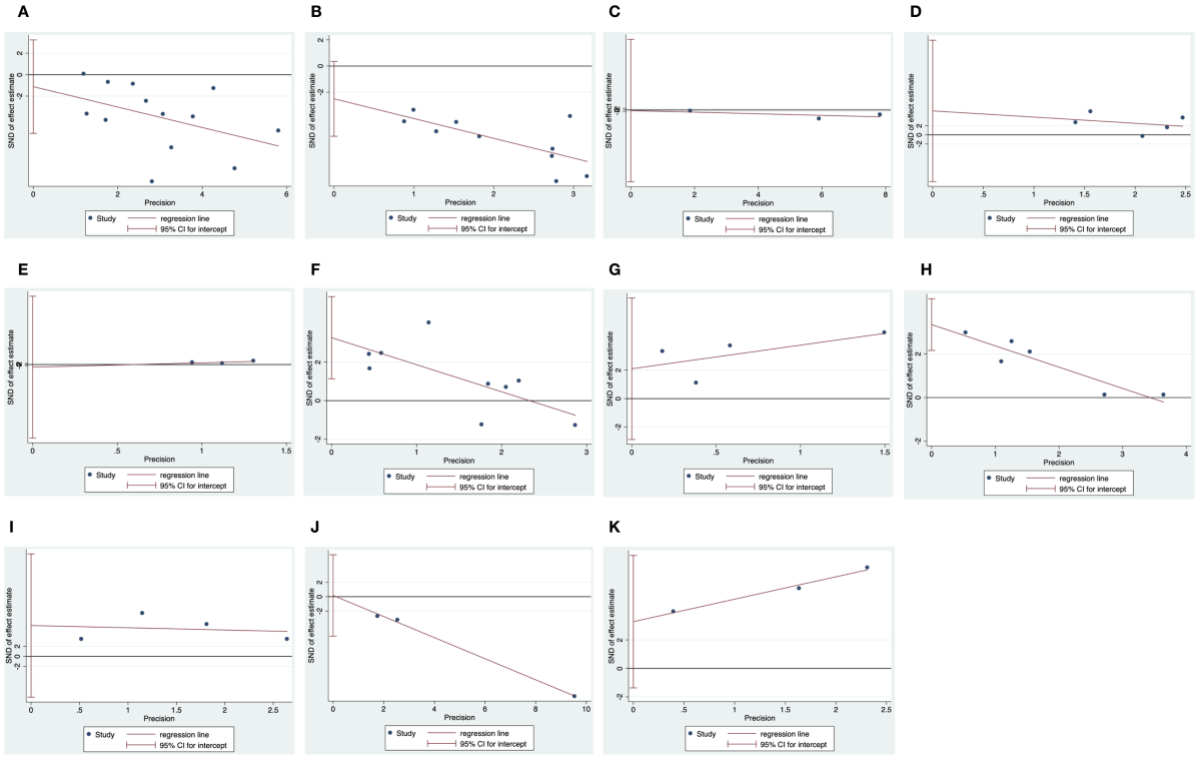
	24	Estimates of diagnostic accuracy and their precision (such as 95% confidence intervals)	6-13
	25	Any adverse events from performing the index test or the reference standard	6-13
DISCUSSION			
	26	Study limitations, including sources of potential bias, statistical uncertainty, and generalisability	17
	27	Implications for practice, including the intended use and clinical role of the index test	17
OTHER INFORMATION			
	28	Registration number and name of registry	N/A
	29	Where the full study protocol can be accessed	N/A
	30	Sources of funding and other support; role of funders	Title Page and Declaration

5. Supplemental Figure 1: Change in admission neutrophil lymphocyte ratio (NLR) standardised mean difference (SMD) upon omission of individual studies



A: Admission NLR association with Good Functional Outcomes; B: Delayed NLR association with Good Functional Outcomes; C: Admission NLR association with mRS 0-1; D: Admission NLR association with Mortality; E: Delayed NLR association with Mortality; F: Admission NLR association with sICH; G: Delayed NLR association with sICH; H: Admission NLR association with ICH; I: Delayed NLR association with ICH; J: Admission NLR association with ENI; K: Admission NLR association with SAI/SAP

6. Supplemental Figure 2: Graphs of Eggers Test Analyses



A: Admission NLR association with Good Functional Outcomes; B: Delayed NLR association with Good Functional Outcomes; C: Admission NLR association with mRS 0-1; D: Admission NLR association with Mortality; E: Delayed NLR association with Mortality; F: Admission NLR association with sICH; G: Delayed NLR association with sICH; H: Admission NLR association with ICH; I: Delayed NLR association with ICH; J: Admission NLR association with ENI; K: Admission NLR association with SAI/SAP

Supplemental Table 4: Meta-analysis of Observational Studies in Epidemiology (MOOSE) Checklist

Item No	Recommendation	Reported on Page No
Reporting of background should include		
1	Problem definition	1-2
2	Hypothesis statement	2
3	Description of study outcome(s)	3-5
4	Type of exposure or intervention used	3-6
5	Type of study designs used	3-5
6	Study population	3-4
Reporting of search strategy should include		
7	Qualifications of searchers (eg, librarians and investigators)	3-4, Title page
8	Search strategy, including time period included in the synthesis and key words	3-4, Supplementary Information
9	Effort to include all available studies, including contact with authors	3-4
10	Databases and registries searched	3-4
11	Search software used, name and version, including special features used (eg, explosion)	3-4
12	Use of hand searching (eg, reference lists of obtained articles)	3-4
13	List of citations located and those excluded, including justification	3-4, Tables 2-3, Figure 1
14	Method of addressing articles published in languages other than English	3-4
15	Method of handling abstracts and unpublished studies	3-4
16	Description of any contact with authors	-
Reporting of methods should include		
17	Description of relevance or appropriateness of studies assembled for assessing the hypothesis to be tested	3-6
18	Rationale for the selection and coding of data (eg, sound clinical principles or convenience)	3-6
19	Documentation of how data were classified and coded (eg, multiple raters, blinding and interrater reliability)	3-6
20	Assessment of confounding (eg, comparability of cases and controls in studies where appropriate)	3-6, Tables 2-3
21	Assessment of study quality, including blinding of quality assessors, stratification or regression on possible predictors of study results	4-5, Supplemental Table 1
22	Assessment of heterogeneity	6-12, Figures 2-5
23	Description of statistical methods (eg, complete description of fixed or random effects models, justification of whether the chosen models account for predictors of study results, dose-response models, or cumulative meta-analysis) in sufficient detail to be replicated	6
24	Provision of appropriate tables and graphics	Tables 1-3, Figures 2-5
Reporting of results should include		

25	Graphic summarizing individual study estimates and overall estimate	Figures 2-5
26	Table giving descriptive information for each study included	Tables 2-3
27	Results of sensitivity testing (eg, subgroup analysis)	6-13, Figures 2-5
28	Indication of statistical uncertainty of findings	6-13
Reporting of discussion should include		
29	Quantitative assessment of bias (eg, publication bias)	4-5, 6-13, Figure 5, Supplemental Table 1
30	Justification for exclusion (eg, exclusion of non-English language citations)	3-4
31	Assessment of quality of included studies	4-5, Supplemental Table 1
Reporting of conclusions should include		
32	Consideration of alternative explanations for observed results	13-17
33	Generalization of the conclusions (ie, appropriate for the data presented and within the domain of the literature review)	17
34	Guidelines for future research	17
35	Disclosure of funding source	Title Page

From: Stroup DF, Berlin JA, Morton SC, et al, for the Meta-analysis Of Observational Studies in Epidemiology (MOOSE) Group. Meta-analysis of Observational Studies in Epidemiology. A Proposal for Reporting. *JAMA*. 2000;283(15):2008-2012. doi: 10.1001/jama.283.15.2008.

Supplemental Table 5: Summary table of individual variations in symptomatic intracerebral haemorrhage (sICH) and intracerebral haemorrhage (ICH) definitions in each study

Study ID	sICH Definition	ICH Definition
1	ECASS III	N/A
3	ECASS II	ECASS II
4	ECASS III	N/A
6	NS	Heidelberg Bleeding Classification
7	ECASS III	ECASS
8	NS	N/A
9	ECASS II	Heidelberg Bleeding Classification
10	ECASS II	MRI used, classified according to ECASS II
11	N/A	Secondary haemorrhage, within or away from the infarction area, which was evident as hyperdense lesions on repeated CT scans 1 week after IVT.
13	ECASS II	N/A
14	SITS-MOST	N/A
15	SITS-MOST	N/A
16	ECASS III	Susceptibility weighted imaging at 7 days after EVT
20	NS	Heidelberg Bleeding Classification
21	ECASS II	MRI used, classified according to ECASS
23	ECASS II	ECASS II
24	Heidelberg Bleeding Classification	N/A
25	ECASS	N/A
27	Fiorelli's analysis of ECASS Definition	Fiorelli's analysis of ECASS Definition
28	Heidelberg Bleeding Classification	Heidelberg Bleeding Classification
33	SITS-MOST	N/A
34	ECASS	ECASS
35	NS	Fisher

Table providing an overall summary of sICH and ICH definitions from each study where this was applicable.

Abbreviations: sICH=Symptomatic intracerebral haemorrhage; ICH=Intracerebral haemorrhage; ECASS=European Cooperative Acute Stroke Study; NS=Not Specified; N/A=Not Applicable; SITS-MOST=Safe Implementation of Thrombolysis in Stroke-Monitoring Study; NIHSS=National Institute of Health Stroke Scale; MRI=Magnetic Resonance Imaging; CT=Computed Tomography; IVT=Intravenous Thrombolysis; EVT=Endovascular Therapy

Supplemental Table 6: Summary of associations between admission NLR and outcomes

Admission NLR Outcome	Overall	IVT ± EVT	IVT only	EVT ± IVT	All Combinations
GFOs	S	S	S	S	N/A
mRS 0-1	S	S	N/A	N/A	N/A
Mortality	S	S	N/A	NS	N/A
sICH	S	NS	N/A	NS	S
ICH	S	NS	N/A	S	N/A
SR	Meta-analysis not able to be carried out				
SAI/SAP	S	N/A	S	S	N/A
ENI	S	S	S	N/A	N/A
END	Meta-analysis not able to be carried out				
DENI					

Table providing an overall summary of results of association between admission NLR and clinical outcomes from meta-analyses carried out. Outcomes are as defined in text.

Abbreviations: NLR=Neutrophil-Lymphocyte Ratio; IVT=Intravenous Thrombolysis; EVT=Endovascular therapy; GFOs=Good Functional Outcomes; S=Statistically significant; N/A=Not Applicable; mRS=Modified Rankin Scale; NS=Not statistically significant; sICH=Symptomatic intracerebral haemorrhage; ICH=Intracerebral haemorrhage; SR=Successful Recanalization; SAI=Stroke Associated Infection; SAP=Stroke Associated Pneumonia; ENI=Early Neurological Improvement; END=Early Neurological Deterioration; DENI=Dramatic Early Neurological Improvement

Supplemental Table 7: Summary of associations between delayed NLR and outcomes

Delayed NLR Outcome	Overall	IVT ± EVT	IVT only	EVT ± IVT	All Combinations
GFOs	S	S	S	S	N/A
mRS 0-1	Meta-analysis not able to be carried out				
Mortality	S	N/A	S	S	N/A
sICH	S	S	N/A	S	N/A
ICH	S	S	N/A	S	S
SR	Meta-analysis not able to be carried out				
SAI/SAP					
ENI					
END					
DENI					

Table providing an overall summary of results of association between delayed NLR and clinical outcomes from meta-analyses carried out. Outcomes are as defined in text.

Abbreviations: NLR=Neutrophil-Lymphocyte Ratio; IVT=Intravenous Thrombolysis; EVT=Endovascular therapy; GFOs=Good Functional Outcomes; S=Statistically significant; N/A=Not Applicable; mRS=Modified Rankin Scale; sICH=Symptomatic Intracerebral Haemorrhage; ICH=Intracerebral haemorrhage; SR=Successful Recanalization; SAI=Stroke Associated Infection; SAP=Stroke Associated Pneumonia; ENI=Early Neurological Improvement; END=Early Neurological Deterioration; DENI=Dramatic Early Neurological Improvement